Operations Research: the final problem

One problem, eight parts, every topic of Lectures 1–5 and every trick the mock and the problem set use

Operations Research, midterm

2026-11-02. Every number recomputed exactly (`verify_final.py`).

How to use it. Pages 1–2 are the paper: do it in one sitting, about two hours, with the formula sheet only. Then compare with the solutions, which follow the same lettering and end each part with the *Trick* box: the thing that the examiners test and that students miss. Conventions as in the Essentials: maximisation, equation-style tableau (basics on the left), $y^\top = c_B^\top B^{-1}$, reduced cost $r_j = c_j - y^\top A_j$, optimal iff constants ≥ 0 and all $r_j \leq 0$.

The paper

The story. A workshop makes three products P_1, P_2, P_3 in divisible quantities $x_1, x_2, x_3 \geq 0$ (units per day); everything made is sold. Three resources are limited per day: machining 5 h, assembly 11 h, finishing 8 h.

	P_1	P_2	P_3
machining (h/unit)	2	3	1
assembly (h/unit)	4	1	2
finishing (h/unit)	3	4	2
profit (hundreds of MAD/unit)	5	4	3

Part A: model. (A1) Formulate the LP that maximises daily profit. Name the constraints R_1 (machining), R_2 (assembly), R_3 (finishing).

Part B: represent. (B1) Write the equational form with slacks s_1, s_2, s_3, variable order $(x_1, x_2, x_3, s_1, s_2, s_3)$, and give A, b, c. State $\operatorname{rank} A$ and justify it in one line. (B2) How many bases can there be at most? (B3) Is $\hat{x} = (x_1, x_2, x_3) = (2, 0, 1)$ a basic feasible solution? If yes give its basis and say whether it is degenerate. (B4) Is R_3 redundant? Prove your answer.

Part C: from a basis to its tableau. For the ordered basis (x_1, x_3, s_2): (C1) B and B^{-1}. (C2) x_B and z_0. (C3) $y^\top = c_B^\top B^{-1}$. (C4) The reduced costs of x_2, s_1, s_3. (C5) Conclude whether the basis is optimal, quoting the two tests.

Part D: simplex. (D1) Starting from the slack basis, run the simplex with the largest-coefficient entering rule. Show every ratio test and the tableau after each pivot. (D2) Which basis would a different entering rule reach first? Does it matter for the final answer?

Part E: read the final tableau. From the optimal tableau alone, read: (E1) the basis and the full optimal solution in the order $(x_1, x_2, x_3, s_1, s_2, s_3)$; (E2) $z^\star$ and which constraints bind; (E3) B^{-1} (check against C1); (E4) y; (E5) the dual slacks; (E6) whether the optimum is unique and whether it is degenerate, with the reason; (E7) the basic direction of x_2, the ratio test if x_2 is forced in, and the plan and value reached.

Part F: duality. (F1) Write the dual. (F2) Check that y from E4 is dual feasible and that $b^\top y = z^\star$; what theorem then proves optimality of both? (F3) Give the certificate as a nonnegative

combination of the primal rows that bounds z by 13. (F4) Write the complementary slackness table for the pair. (F5) A colleague proposes $\tilde{x} = (\frac{5}{2}, 0, 0)$. Use complementary slackness, not the tableau, to decide whether it is optimal.

Part G: sensitivity. (G1) Machining becomes $5+\delta$: range of δ, $z^\star(\delta)$, plan at $\delta = \frac{1}{3}$. Then $\delta = 1$: show the basis fails, reoptimise, give plan, value and y, and compare with the local formula. (G2) Assembly becomes $11+\delta$: range and shadow price, with the one-line reason. (G3) Finishing becomes $8+\delta$: range and shadow price; what happens exactly at the upper endpoint and beyond it? (G4) Profit of P_1 becomes $5+\varepsilon$: range, value, and the complete set of optimal plans at each endpoint. (G5) Profit of P_2 becomes $4+t$: for which t does the plan change, and what is the break-even profit of P_2? (G6) A new product P_4 uses 1 h of each resource and earns γ: for which γ is it worth making? For $\gamma = 3$ find all optimal plans. (G7) Finishing becomes $8+\delta$ and the profit of P_1 becomes $5+\varepsilon$ at the same time: region where the basis stays optimal, and $z^\star(\delta, \varepsilon)$. Is it additive?

Part H: theory and traps. (H1) The true profits are only known to within $\pm\frac{1}{2}$ of the table. Prove that the true optimal value v satisfies $11 \le v \le 15$. (H2) Show by an explicit profit vector within $\pm\frac{1}{2}$ that the plan $(2, 0, 1)$ is *not* guaranteed to stay optimal. (H3) Finishing drops to 7.5 h. Show that the optimal vertex is degenerate, give two bases that describe it and their two multiplier vectors, and say what this means for "the" shadow price of finishing. (H4) True or false, one line each: (i) a nonbasic reduced cost of 0 at an optimal tableau always means several optimal solutions; (ii) if a slack is basic its resource has shadow price 0; (iii) an unbounded feasible region makes the LP unbounded; (iv) after a resource change inside the range the optimal plan is unchanged; (v) $z^\star(b)$ is convex in b.

Solutions and tricks

Part A: model

Solution

(A1) $\max z = 5x_1 + 4x_2 + 3x_3$ s.t. $R_1: \; 2x_1 + 3x_2 + x_3 \le 5$, $R_2: \; 4x_1 + x_2 + 2x_3 \le 11$, $R_3: \; 3x_1 + 4x_2 + 2x_3 \le 8$, $x_1, x_2, x_3 \ge 0$. Units: hours per day on the left, hours available on the right; profit in hundreds of MAD per day.

Trick

Say what the variables are (units per day, divisible) and keep the units of each row consistent. "Everything made is sold" is what makes profit linear in x.

Part B: represent

Solution

(B1) $\max c^\top x$ s.t. $Ax = b$, $x \ge 0$ with

$$A = \begin{bmatrix} 2 & 3 & 1 & 1 & 0 & 0 \\ 4 & 1 & 2 & 0 & 1 & 0 \\ 3 & 4 & 2 & 0 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 11 \\ 8 \end{bmatrix}, \quad c = (5, 4, 3, 0, 0, 0)^\top.$$

$\operatorname{rank} A = 3$: the three slack columns form I_3.
(B2) At most $\binom{6}{3} = 20$ bases (fewer if some 3×3 submatrix is singular). Only six are feasible here, but you are not asked to list them.
(B3) At $\hat{x} = (2, 0, 1)$: R_1: $4 + 1 = 5$, tight, $s_1 = 0$. R_2: $8 + 2 = 10 < 11$, $s_2 = 1$. R_3: $6 + 2 = 8$, tight, $s_3 = 0$. Full vector $(2, 0, 1, 0, 1, 0)$: three positive entries x_1, x_3, s_2, and their columns $(2, 4, 3), (1, 2, 2), (0, 1, 0)$ are independent (the determinant of B in C1 is $-1 \neq 0$). So $\hat{x}$ is a BFS with basis (x_1, x_3, s_2), and it is **non-degenerate** (all three basic values positive).
(B4) Not redundant. $x = (0, 0, 5)$ satisfies R_1 ($5 \le 5$) and R_2 ($10 \le 11$) but violates R_3 ($10 > 8$). A redundant row never cuts the region defined by the others.

Trick

Redundancy is about the *region*, not about the optimum: a row can be slack at the optimum and still be non-redundant (R_2 here). To prove non-redundancy, exhibit one point; to prove redundancy, write the row as a nonnegative combination of the others plus $x \ge 0$.

Part C: from a basis to its tableau

Solution

(C1) Columns of x_1, x_3, s_2 in that order:

$$B = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 2 & 1 \\ 3 & 2 & 0 \end{bmatrix}, \qquad B^{-1} = \begin{bmatrix} 2 & 0 & -1 \\ -3 & 0 & 2 \\ -2 & 1 & 0 \end{bmatrix}.$$

Check $BB^{-1} = I$: row 1 of B times column 1 of B^{-1}: $4 - 3 + 0 = 1$; times column 3: $-2 + 2 = 0$.
(C2) $x_B = B^{-1}b = (10 - 8, \; -15 + 16, \; -10 + 11) = (2, 1, 1)$: $x_1 = 2$, $x_3 = 1$, $s_2 = 1$. $z_0 = 5 \cdot 2 + 3 \cdot 1 + 0 = 13$.
(C3) $c_B = (5, 3, 0)$; $y^\top = c_B^\top B^{-1} = (10 - 9, \; 0, \; -5 + 6) = (1, 0, 1)$.

(C4) $r_{x_2} = 4 - (1 \cdot 3 + 0 + 1 \cdot 4) = -3$; $r_{s_1} = 0 - y_1 = -1$; $r_{s_3} = 0 - y_3 = -1$.
(C5) Feasibility $x_B = (2, 1, 1) \geq 0$ and optimality $r = (-3, -1, -1) \leq 0$: both tests hold, the basis is optimal with $z^\star = 13$. Displayed columns $-B^{-1}A_j$: x_2: $-(6-4,\ -9+8,\ -6+1) = (-2, 1, 5)$; s_1: $-(2, -3, -2)$; s_3: $-(-1, 2, 0)$. So the tableau is

$$x_1 = 2 - 2x_2 - 2s_1 + s_3, \quad x_3 = 1 + x_2 + 3s_1 - 2s_3, \quad s_2 = 1 + 5x_2 + 2s_1, \quad z = 13 - 3x_2 - s_1 - s_3.$$

Trick

Order matters twice: the columns of B follow the *ordered* basis, and c_B follows the same order. A sign error in $-B^{-1}A_j$ flips the whole tableau; check one row against the original equation ($x_1 = 2 - 2x_2 - 2s_1 + s_3$ with everything nonbasic $= 0$ gives R_1: $4 + 0 + 1 = 5$, correct).

Part D: simplex

Solution

(D1) Start: $s_1 = 5 - 2x_1 - 3x_2 - x_3$, $s_2 = 11 - 4x_1 - x_2 - 2x_3$, $s_3 = 8 - 3x_1 - 4x_2 - 2x_3$, $z = 5x_1 + 4x_2 + 3x_3$.
Pivot 1: x_1 enters (coefficient 5). Ratios 5/2, 11/4, 8/3: s_1 leaves at $x_1 = \frac{5}{2}$. Solve R_1: $x_1 = \frac{5}{2} - \frac{3}{2}x_2 - \frac{1}{2}x_3 - \frac{1}{2}s_1$; substitute:

$$s_2 = 1 + 5x_2 + 2s_1, \quad s_3 = \tfrac{1}{2} + \tfrac{1}{2}x_2 - \tfrac{1}{2}x_3 + \tfrac{3}{2}s_1, \quad z = \tfrac{25}{2} - \tfrac{7}{2}x_2 + \tfrac{1}{2}x_3 - \tfrac{5}{2}s_1.$$

Pivot 2: only x_3 has a positive coefficient. Ratios: row x_1: $\frac{5}{2}/\frac{1}{2} = 5$; row s_2: coefficient 0, no limit; row s_3: $\frac{1}{2}/\frac{1}{2} = 1$. s_3 leaves at $x_3 = 1$. Solve: $x_3 = 1 + x_2 + 3s_1 - 2s_3$; substitute:

$$x_1 = 2 - 2x_2 - 2s_1 + s_3, \quad s_2 = 1 + 5x_2 + 2s_1, \quad z = 13 - 3x_2 - s_1 - s_3.$$

All z-row coefficients ≤ 0: optimal, $(x_1, x_2, x_3) = (2, 0, 1)$, $z^\star = 13$. Same tableau as C5.
(D2) With the smallest-index or "largest increase" rule the first pivot could bring in x_3 (ratios 5, $\frac{11}{2}$, 4: s_3 leaves, value 12) and reach the basis (x_3, s_1, s_2) first. The path changes, the optimum does not: every rule stops at a tableau passing both tests, and here the optimum is unique (E6).

Trick

A row whose coefficient of the entering variable is ≥ 0 never limits the step (row s_2 in pivot 2): leave it out of the ratio test. The objective after pivot 1 is $\frac{25}{2}$, after pivot 2 it is 13: it must increase strictly at every non-degenerate pivot, which is your running check.

Part E: read the final tableau

Solution

(E1) Basis (x_1, x_3, s_2) (left-hand sides). Full solution $(x_1, x_2, x_3, s_1, s_2, s_3) = (2, 0, 1, 0, 1, 0)$.
(E2) $z^\star = 13$. Binding: R_1 and R_3 (their slacks are nonbasic, $= 0$); R_2 has 1 h spare.
(E3) B^{-1} = minus the slack columns, with a unit column for the basic slack s_2: column of s_1 is $(-2, 3, 2)^\top$ so $B^{-1}e_1 = (2, -3, -2)^\top$; $B^{-1}e_2 = e_2$ (the position of s_2 in the basis, second row); column of s_3 is $(1, -2, 0)^\top$ so $B^{-1}e_3 = (-1, 2, 0)^\top$. Same matrix as C1.
(E4) $y = -(r_{s_1}, r_{s_2}, r_{s_3}) = (1, 0, 1)$ ($r_{s_2} = 0$ since s_2 is basic).
(E5) Dual slack of the dual constraint of x_j is $-r_j$: $(0, 3, 0)$ for (x_1, x_2, x_3).
(E6) Unique: every nonbasic reduced cost is strictly negative $(-3, -1, -1)$, so any other feasible point with some nonbasic > 0 has $z < 13$. Non-degenerate: constants $2, 1, 1$ all positive.
(E7) Direction of x_2: read its column, (x_1, x_3, s_2) change by $(-2, +1, +5)$ per unit, z by -3. Ratio

test: only x_1 decreases, $2/2 = 1$: $x_2 = 1$, x_1 leaves. Plan $(0, 1, 2)$, slacks $(0, 6, 0)$, $z = 13 - 3 = 10$.

Trick

The basic slack gives a *unit column* of B^{-1} in the position of that slack in the basis, not a column of zeros. Dual slacks are $-r_j$, dual values are $-r_{s_i}$: both are minus signs in this layout, plus signs in the z-row layout of other books.

Part F: duality

Solution

(F1) min $w = 5y_1 + 11y_2 + 8y_3$ s.t. $2y_1 + 4y_2 + 3y_3 \geq 5$ (x_1), $3y_1 + y_2 + 4y_3 \geq 4$ (x_2), $y_1 + 2y_2 + 2y_3 \geq 3$ (x_3), $y \geq 0$.
(F2) $y = (1, 0, 1)$: $2 + 3 = 5 \geq 5$ tight; $3 + 4 = 7 \geq 4$ slack 3; $1 + 2 = 3 \geq 3$ tight; $y \geq 0$. $w = 5 + 8 = 13 = z^\star$. Weak duality: a feasible pair with equal values is optimal on both sides (no need for strong duality here).
(F3) $1 \cdot R_1 + 1 \cdot R_3$: $5x_1 + 7x_2 + 3x_3 \leq 13$. For $x \geq 0$, $z = 5x_1 + 4x_2 + 3x_3 \leq 5x_1 + 7x_2 + 3x_3 \leq 13$. Attained at $(2, 0, 1)$.
(F4) Complementary slackness:

pair	primal side	dual side	product
R_1, y_1	tight $(s_1 = 0)$	$y_1 = 1$	0
R_2, y_2	slack 1	$y_2 = 0$	0
R_3, y_3	tight	$y_3 = 1$	0
x_1, dual row 1	$x_1 = 2$	tight	0
x_2, dual row 2	$x_2 = 0$	slack 3	0
x_3, dual row 3	$x_3 = 1$	tight	0

(F5) $\tilde{x} = (\frac{5}{2}, 0, 0)$: R_1 tight (5), R_2 slack $(10 < 11)$, R_3 slack $(7.5 < 8)$. CS forces $y_2 = y_3 = 0$. $x_1 > 0$ forces dual row 1 tight: $2y_1 = 5$, $y_1 = \frac{5}{2}$. Check dual feasibility: row 2: $\frac{15}{2} \geq 4$ ok; row 3: $\frac{5}{2} \geq 3$ **fails**. No dual solution satisfies CS with $\tilde{x}$, so $\tilde{x}$ is not optimal. (Indeed $z(\tilde{x}) = 12.5 < 13$.)

Trick

CS is a *test*, not a computation of the optimum: tight primal rows give the dual unknowns, positive primal variables give the equations, and the last step, checking the remaining dual inequalities, is the one that decides. Forgetting it is the classic error.

Part G: sensitivity

Solution

(G1) $g = B^{-1}e_1 = -(\text{column of } s_1) = (2, -3, -2)^\top$. Constants $2 + 2\delta \geq 0$, $1 - 3\delta \geq 0$, $1 - 2\delta \geq 0$: $-1 \leq \delta \leq \frac{1}{3}$. $z^\star(\delta) = 13 + \delta$ $(y_1 = 1)$. At $\delta = \frac{1}{3}$: plan $(x_1, x_3, s_2) = (\frac{8}{3}, 0, \frac{1}{3})$, $z = \frac{40}{3}$, degenerate (the endpoint).
$\delta = 1$: constants $(4, -2, -1)$: $x_3 < 0$ and $s_2 < 0$, basis infeasible, reduced costs still $(-3, -1, -1)$. Dual simplex on the most negative row $x_3 = -2 + x_2 + 3s_1 - 2s_3$: positive coefficients x_2 (ratio 3/1) and s_1 (ratio 1/3): s_1 enters, x_3 leaves. $s_1 = \frac{2}{3} - \frac{1}{3}x_2 + \frac{1}{3}x_3 + \frac{2}{3}s_3$; substitute:

$$x_1 = \tfrac{8}{3} - \tfrac{4}{3}x_2 - \tfrac{2}{3}x_3 - \tfrac{1}{3}s_3, \quad s_2 = \tfrac{1}{3} + \tfrac{13}{3}x_2 + \tfrac{2}{3}x_3 + \tfrac{4}{3}s_3, \quad z = \tfrac{40}{3} - \tfrac{8}{3}x_2 - \tfrac{1}{3}x_3 - \tfrac{5}{3}s_3.$$

All constants ≥ 0: optimal. Plan $(\frac{8}{3}, 0, 0)$, $z^\star = \frac{40}{3} \approx 13.33$, $y = (0, 0, \frac{5}{3})$: machining is no longer binding $(s_1 = \frac{2}{3})$, its price fell to 0; finishing's price rose. The local formula would give

$13+1=14$: wrong beyond the range, and above the truth (concavity).
(G2) s_2 is basic: $g=e_2$, the only condition is $1+\delta\geq 0$: $\delta\geq -1$, no upper limit, $y_2=0$. Reason: assembly has 1 h spare, an extra hour is worthless and up to 1 h can be removed for free.
(G3) $g=-(\text{column of } s_3)=(-1,2,0)^\top$. $2-\delta\geq 0$, $1+2\delta\geq 0$: $-\frac{1}{2}\leq\delta\leq 2$, $y_3=1$, $z^\star=13+\delta$. At $\delta=2$: plan $(0,0,5)$, $x_1=0$ while basic: degenerate; two bases describe this vertex and $z=15$. Beyond $\delta>2$: $x_1<0$, dual simplex ($x_1=-2x_2-2s_1+s_3+(2-\delta)$, s_3 enters), new basis (x_3,s_2,s_3) with $y=(3,0,0)$, $z^\star=15$ for all $\delta\geq 2$: the slope dropped from 1 to 0 because machining alone now limits $x_3\leq 5$.
(G4) Add $\varepsilon\times$ (row x_1) to the z row: $z=13+2\varepsilon+(-3-2\varepsilon)x_2+(-1-2\varepsilon)s_1+(-1+\varepsilon)s_3$. Conditions $\varepsilon\geq-\frac{3}{2}$, $\varepsilon\geq-\frac{1}{2}$, $\varepsilon\leq 1$: $-\frac{1}{2}\leq\varepsilon\leq 1$, i.e. $c_1\in[4.5,6]$; plan $(2,0,1)$ unchanged, $z^\star=13+2\varepsilon$, $y=(1+2\varepsilon,0,1-\varepsilon)$.
Endpoint $\varepsilon=1$: $r_{s_3}=0$. Let s_3 enter: ratios row x_1 (coefficient $+1$, no), row x_3 ($1/2$), row s_2 (0, no): $s_3=\frac{1}{2}$, x_3 leaves, plan $(\frac{5}{2},0,0)$. All optima: the segment $(2,0,1)+\theta(\frac{1}{2},0,-1)$, $0\leq\theta\leq 1$, value 15.
Endpoint $\varepsilon=-\frac{1}{2}$: $r_{s_1}=0$; s_1 enters, ratios row x_1 ($2/2=1$), others increase: $s_1=1$, x_1 leaves, plan $(0,0,4)$. All optima: $(2,0,1)+\theta(-2,0,3)$, $0\leq\theta\leq 1$, value 12.
(G5) x_2 nonbasic: $r_2=-3\to -3+t$. Plan unchanged iff $t\leq 3$. Break-even profit $7=y^\top A_2=3+4$: at profit 7 the plan $(0,1,2)$ of E7 (value $7+6=13$) ties with $(2,0,1)$; above 7, P_2 enters.
(G6) $a=(1,1,1)^\top$, $y^\top a=2$, $r_u=\gamma-2$: worth making iff $\gamma>2$. For $\gamma=3$: $r_u=1$, column $-B^{-1}a=-(1,-1,-1)=(-1,1,1)$: $x_1=2-u$, $x_3=1+u$, $s_2=1+u$. Only x_1 limits: $u=2$, x_1 leaves. New tableau

$$u=2-x_1-2x_2-2s_1+s_3,\qquad x_3=3-x_1-x_2+s_1-s_3,$$
$$s_2=3-x_1+3x_2+s_3,\qquad z=15-x_1-5x_2-3s_1+0\,s_3.$$

Optimal, $z^\star=15$, plan $(x_1,x_2,x_3,u)=(0,0,3,2)$, $y=(3,0,0)$. $r_{s_3}=0$: let s_3 enter, ratio row x_3: $3/1=3$: second optimal vertex $(0,0,0,5)$ (make only P_4). All optima: $(0,0,3,2)+\theta(0,0,-3,3)$, $0\leq\theta\leq 1$.
(G7) Feasibility depends on δ only: $-\frac{1}{2}\leq\delta\leq 2$. Optimality on ε only: $-\frac{1}{2}\leq\varepsilon\leq 1$. Rectangle. Value $c_B(\varepsilon)^\top(x_B+\delta g)=(5+\varepsilon)(2-\delta)+3(1+2\delta)=13+\delta+2\varepsilon-\varepsilon\delta$. Not additive: the cross term $-\varepsilon\delta$ is the extra profit per unit of P_1 times the *loss* of P_1 output caused by more finishing (more finishing shifts production from P_1 to P_3).

Trick

G1: two negative constants, pivot on one row at a time; the ratio uses $-r_j$ over the *positive* coefficients of that row. G2: basic slack, price 0, one-sided range, no computation. G3: at the endpoint the vertex is degenerate and beyond it the slope drops, often to 0. G4: a zero reduced cost at an endpoint is an invitation to pivot and describe the whole optimal segment. G6: zero reduced cost after the pivot, so "find all optimal plans" needs a second pivot. G7: the cross term can be negative.

Part H: theory and traps

Solution

(H1) Let d be the perturbation, $|d_j|\leq\frac{1}{2}$, and P the feasible region (unchanged). From R_3: $2(x_1+x_2+x_3)\leq 3x_1+4x_2+2x_3\leq 8$, so $x_1+x_2+x_3\leq 4$ on P and $|d^\top x|\leq\frac{1}{2}\cdot 4=2=:\sigma$. With $x^\star=(2,0,1)$ optimal for c and $\tilde{x}$ optimal for $c+d$, both in P: $v\geq(c+d)^\top x^\star=13+d^\top x^\star\geq 11$, and $v=c^\top\tilde{x}+d^\top\tilde{x}\leq 13+2=15$. (Exact extremes are 11.5 and 14.5, the bound is not tight; it is not meant to be.)
(H2) Take profits $(4.5,4,3.5)$, i.e. $\varepsilon_1=-\frac{1}{2}$, $\varepsilon_3=+\frac{1}{2}$. New z row = old $+\varepsilon_1$(row x_1)$+\varepsilon_3$(row x_3):

coefficient of s_1 is $-1-2\varepsilon_1+3\varepsilon_3 = -1+1+\frac{3}{2} = \frac{3}{2} > 0$. The optimality test fails, s_1 enters, and the new optimum is $(0,0,4)$ with value 14 (whereas $(2,0,1)$ gives 12.5). So H1 bounds the *value*; it says nothing about the *plan*.

(H3) $b_3 = 7.5$, $\delta = -\frac{1}{2}$, the lower endpoint of G3: constants $x_B + \delta g = (2.5, 0, 1)$. $x_3 = 0$ while basic: degenerate vertex $(2.5, 0, 0)$, $z = 12.5$. A second basis for the same point: (x_1, s_1, s_2), with $x_B = (\frac{5}{2}, 0, 1)$ (now s_1 is the basic variable at 0) and multipliers $y = (0, 0, \frac{5}{3})$, against $y = (1, 0, 1)$ for (x_1, x_3, s_2). Both multiplier vectors are dual optimal: $b^\top y = 5 + 7.5 = 12.5$ and $\frac{5}{3} \cdot 7.5 = 12.5$. So the dual has several optima and finishing has no single shadow price: the value function has a kink at $b_3 = 7.5$, slope 1 to the right (basis (x_1, x_3, s_2), valid up to $b_3 = 10$) and slope $\frac{5}{3}$ to the left (basis (x_1, s_1, s_2), valid for $b_3 \leq 7.5$).

(H4) (i) False: at a degenerate optimum a zero reduced cost may give a step of length 0; true if the tableau is non-degenerate. (ii) True: $y_i = -r_{s_i}$ and a basic variable has reduced cost 0. (iii) False: $\max -x_1$ on $x_1 \geq 0$. (iv) False: the *basis* is unchanged, the plan is $x_B + \delta g$. (v) False: concave (minimum of the linear functions $b^\top y$ over the dual feasible set).

Trick

H1: name the bounded region and where σ comes from; then two one-line inequalities, each using that the optimum of one problem is feasible for the other. H2: "value within σ" never implies "same plan". H3: degenerate primal vertex $\Leftrightarrow$ non-unique dual optimum $\Leftrightarrow$ no single shadow price; the one-sided slopes are the two multiplier vectors. H4(iii): unbounded region versus unbounded objective is a favourite true/false.